

Fine-Grained Emotion Structure in the Brain-Aligned Subspace of a Self-Supervised Video Model

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Abstract

Do the brain and self-supervised video models share a common structure for representing emotion? A central debate in affective neuroscience concerns whether emotions are represented along affective dimensions such as valence and arousal, or as fine-grained emotion categories. Prior work showed that neural responses to emotional videos are better explained by fine-grained emotion categories than by affective dimensions. Here, we ask whether this categorical structure extends to the subspace shared between the human brain and a video foundation model trained without emotion labels. By aligning neural representations from a brain foundation model (Brain-JEPA) with embeddings from a self-supervised video foundation model (V-JEPA2), we identified a compact brain-aligned subspace restricted to only three of the leading principal components of a 1,408-dimensional embedding. This shared neural-computational subspace shows stronger geometric alignment with discrete emotion categories than with affective dimensions, with categorical dominance exceeding that of the full video model space, a pattern stable across all individual subjects. These findings suggest that the visual features shared between a self-supervised video model and the subject-invariant brain response organize emotional videos along categorical rather than dimensional lines, and that the brain's dimensional (valence–arousal) structure for emotional video resides predominantly outside this visually shared channel.

Keywords: emotion representation; fMRI; video foundation model; self-supervised learning; brain-aligned subspace

Introduction

A central question in affective neuroscience concerns how the brain organizes emotional experience, whether along continuous affective dimensions such as valence and arousal (Russell, 1980), or as a high-dimensional semantic space of fine-grained emotion categories (Cowen & Keltner, 2017). To approach this question, Horikawa et al. (2020) collected whole-brain fMRI responses while subjects viewed over 2,000 emotionally evocative videos, and demonstrated that neural responses are better predicted by fine-grained emotion

categories than by affective dimensions, with emotion-related representations distributed across transmodal brain regions. These findings suggest that the brain represents emotion in a high-dimensional, categorically organized geometry. Recent work at the intersection of AI and affective neuroscience has shown that artificial neural networks can successfully predict human emotional responses to visual stimuli, suggesting that emotion schemas are deeply embedded within both biological and artificial visual systems (Conwell et al., 2025; Kragel et al., 2019). However, these studies have primarily relied on supervised models trained with explicit conceptual or object categories, or on behavioral rather than direct neural measurements. In parallel, self-supervised video foundation models learn structured visual representations from large-scale video data without any semantic or affective supervision. Here we examine the affective geometry of the subspace shared between such models and the brain, by aligning neural representations from Brain-JEPA (Dong et al., 2024) with embeddings from V-JEPA2 (Assran et al., 2025), a self-supervised video foundation model trained on over one million hours of video. We find that the brain-aligned subspace of V-JEPA2 is compact and shows stronger geometric alignment with categorical than dimensional emotion structure.

Methods

We used the fMRI dataset from Horikawa et al. (2020), in which five healthy adult subjects each viewed 2,196 emotionally evocative short videos. Emotion ratings were obtained from independent crowd workers, yielding continuous scores for 34 emotion categories and 14 affective dimension ratings (Cowen & Keltner, 2017). Neural representations were extracted per subject by passing preprocessed fMRI through Brain-JEPA (Dong et al., 2024), a self-supervised fMRI foundation model pretrained on UK Biobank data, without task-specific fine-tuning. Subject-level embeddings were averaged across subjects for group-level analyses. Video representations were obtained from V-JEPA2 (Assran et al., 2025) by uniformly sampling 16 frames per video and mean-pooling the final hidden state across tokens, yield-



ing a (2196, 1408) embedding. To identify which dimensions of V-JEPA2 covary with brain activity, we first decomposed the video embeddings into 100 principal components (PCs) to capture the majority of variance, and predicted each PC from the group-mean Brain-JEPA representation via 5-fold cross-validated ridge regression. PCs surviving permutation-based FDR correction ($q < 0.05$, $n = 1,000$ permutations) were defined as brain-aligned. We then examined the affective structure of this brain-aligned subspace. Using the same regression pipeline, we decoded 34 emotion category scores and, among the 14 affective dimensions, focused on valence and arousal (V-A) as the two canonical axes of the dimensional model of emotion. Subject-level stability was assessed by repeating the brain-alignment and decoding analyses for each subject individually, and a one-sided Wilcoxon signed-rank test was applied to the per-subject category/V-A ratios against the null of ratio ≤ 1 . To characterize the affective geometry of the brain-aligned subspace, we computed the ratio of mean category R^2 to mean V-A R^2 (category/V-A ratio). We computed the same ratio for the full video model space (all 100 PCs) as a baseline comparison. To control for low-level visual and semantic confounds, VGG19-based vision features (1,000-dim) and human-annotated semantic features (73-dim) were regressed out from both representations and targets, and partial R^2 was recomputed. As a robustness check, the brain-alignment pipeline was repeated with two alternative brain-side representations (an untrained Brain-JEPA with random initialization and 450-parcel raw BOLD averaged across subjects), and additionally in the reverse direction with both representations reduced to 100 PCs.

Results

Of 100 principal components, only three survived permutation-based FDR correction (PC1: $R^2 = 0.373$, PC2: $R^2 = 0.075$, PC3: $R^2 = 0.088$), while all remaining components fell near zero. This indicates a highly compressed shared space, dominated by PC1 with PC2 and PC3 contributing substantially smaller effects, and anchored by PC1 which remained stable across all five individual subjects. These brain-aligned components showed substantially stronger maximum absolute Spearman correlations with emotion categories than brain-unaligned components (mean max $|r| = 0.33$ vs. 0.09, FDR-corrected), confirming that the shared neural-computational subspace is affectively organized (Figure 1).

Decoding from this brain-aligned subspace revealed stronger prediction of emotion categories than valence and arousal (mean $R^2 = 0.055$ vs. 0.038, category/V-A ratio = 1.44). This categorical dominance exceeded that of the full V-JEPA2 space (ratio = 1.26), indicating that the dimensions selectively shared with the brain are

more categorically organized than V-JEPA2’s full representational space. This pattern was stable across all five subjects, with per-subject ratios ranging from 1.96 to 2.65 (all > 1 ; one-sided Wilcoxon signed-rank $p = 0.031$). After regressing out low-level vision and semantic features, absolute predictive values decreased substantially, though the categorical organization of the shared subspace was attenuated but preserved (Figure 2).

Discussion

Our findings reveal that the representational space shared between a self-supervised video foundation model and the human brain is remarkably compact: only three of the leading 100 principal components of V-JEPA2 survive permutation-based FDR correction, and this subspace is dominated by PC1 ($R^2 = 0.373$) with PC2 and PC3 ($R^2 = 0.075$ and 0.088) contributing substantially smaller effects.

Inside this brain-aligned subspace, the geometric organization of emotional videos aligns more strongly with discrete emotion categories than with continuous valence–arousal dimensions. The category-to-dimension ratio is 1.44 inside the brain-aligned subspace versus 1.26 in the full V-JEPA2 100-PC space, and per-subject ratios range from 1.96 to 2.65 across all five participants (all > 1 , $p = 0.031$). This pattern parallels the categorical dominance observed by Horikawa et al. (2020) in raw fMRI responses to the same stimulus set, while extending it to a label-free, self-supervised foundation-model framing.

Repeating the analysis with two alternative brain-side encoders (untrained Brain-JEPA and raw 450-parcel BOLD) and in the reverse direction preserved this categorical asymmetry. Although the brain’s full representational space favored dimensional over categorical decoding (full-space ratios 0.60–0.76), its V-JEPA2-aligned subspace reversed toward categorical dominance (top-3 ratios 1.05–1.42), suggesting that the shared channel isolates the visual, category-organized contribution to brain emotion processing while the brain’s dimensional structure resides predominantly outside it. This split is consistent with constructionist accounts of emotion in which sensory ingredients contribute discriminating category structure while non-sensory ingredients contribute continuous affective dimensions.

Two extensions follow naturally. First, comparison with visual-language foundation models (e.g., CLIP) will test whether language-mediated visual learning sharpens the categorical channel further (Doerig et al., 2025). Second, decomposing the non-visual sources of dimensional structure (e.g., interoceptive, semantic, or memory-related brain networks) would test whether the categorical-versus-dimensional split between visual and

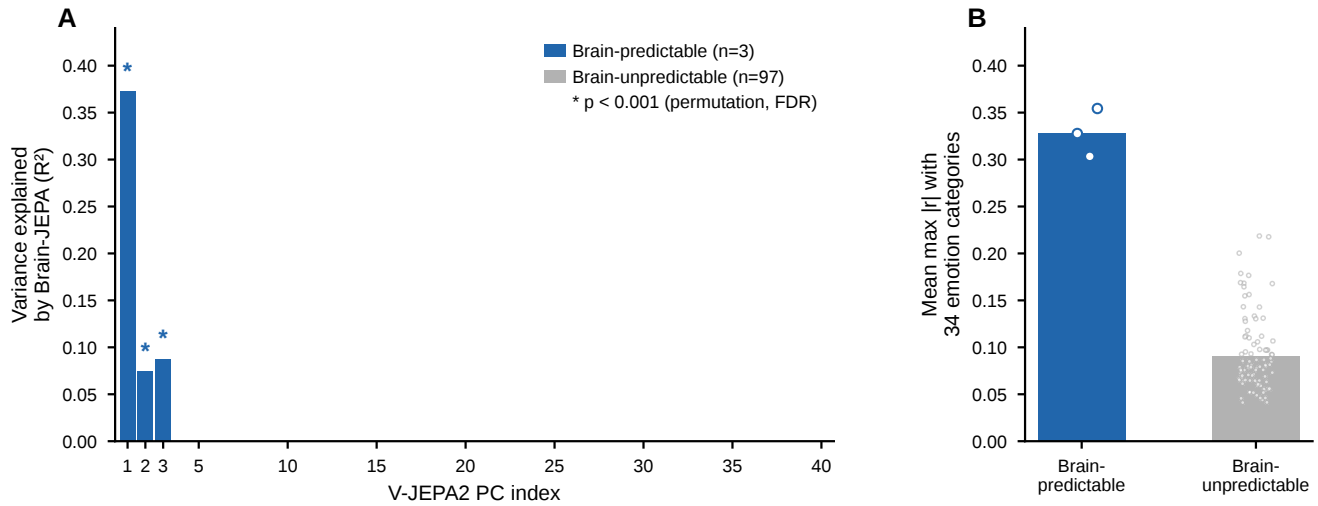


Figure 1: Brain-aligned subspace of V-JEPA2. (A) R^2 for predicting each V-JEPA2 PC from Brain-JEPA. Only PC1–3 survived permutation-based FDR correction. (B) Mean max $|r|$ with emotion categories for brain-aligned vs. brain-unaligned PCs (0.33 vs. 0.09).

non-visual contributions generalizes across paradigms.

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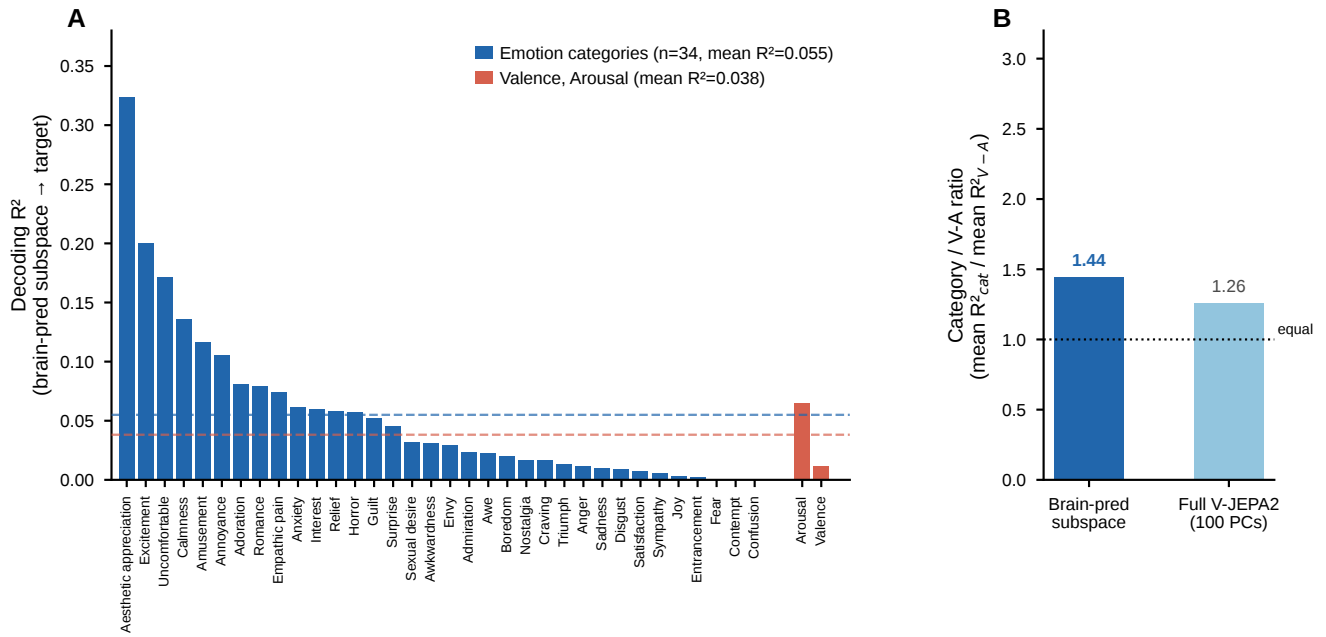


Figure 2: Categorical emotion structure of the brain-aligned subspace. (A) Emotion decoding R^2 from the brain-aligned subspace across 34 categories. Valence and Arousal shown separately. (B) category/V-A ratio for the brain-aligned subspace (1.44) vs. the full V-JEPA2 space (1.26). Ratio > 1 in all five subjects.